

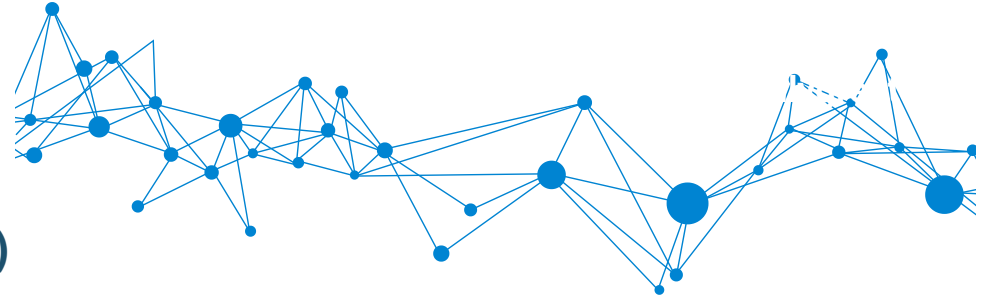
Assignment Feedback!

I enjoy getting feedback from my professors. It's me know where my opportunities are to improve, and encourages me. The Professor that teaches **Introduction to Network Concepts** that I am taking this semester gives really good detailed feedback. Last weeks class discussion we were asked to use AI to help troubleshoot a Local Area Network (LAN) issue. In the following slides I provide the assignment instructions, the steps I took, results, and my professors feedback.



DISCUSSION:

Using Generative AI to Troubleshoot Local Area Networks (LAN)



Generative AI Tool

Microsoft CoPilot

Craft AI Prompt

Objective, Context, Technical
Details, Specifications or
Constraints

Reflection

How the tool could help these
kinds of issue, benefits using AI,
limitations, risks, challenges
when relying on AI

AI Prompt: I need to troubleshoot a Local Area Networking (LAN) issue with network connectivity issues for a class assignment with a practice scenario and/or simulated workplace problem. Include relevant information such as network size, device types, wired/wireless setup, IP version (IPv4 or IPv6), and tools or protocols involved. Boundaries for the task: small office LAN, focus on wired connections only.

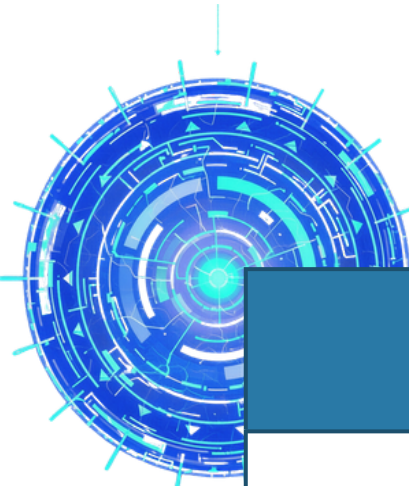


Network Description

Scenario Overview: A small office with 25 employees reports that several workstations cannot access internal resources (file server, printer, VoIP desk phones) and have intermittent Internet connectivity. The issue appears isolated to one side of the office. The LAN is wired-only, using Ethernet cabling, managed switches, and IPv4 addressing. No wireless access points are involved.

Network Size	Device Types	Wired Setup	IP Version
• ~25 wired workstations • 3 network printers • 1 VoIP phone system • 1 file server • 2 managed switches • 1 router/firewall connecting LAN → ISP	• Windows 10/11 desktops • VoIP desk phones • Network printers • Managed switches (Layer 2) • Router/firewall (Layer 3)	• Cat6 cabling • Patch panels → switches → desks • Switches connected via uplink ports • No Wi-Fi in scope	• IPv4 • Static IPs for server/printers • DHCP for workstations • Subnet: 192.168.10.0/24





Tools & Protocols Used and Troubleshooting



Tools & Protocols Used

- Ping (ICMP)
- ipconfig / ifconfig
- tracert / traceroute
- ARP table inspection
- Switch port status (CLI or web UI)
- Cable tester
- DHCP protocol
- Ethernet (802.3)
- TCP/UDP connectivity tests

Practical Troubleshooting Scenario

Problem Statement

Employees in the east wing report:

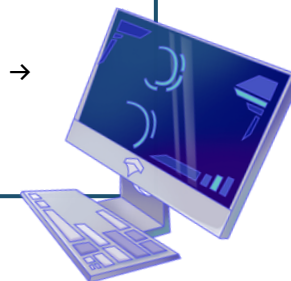
- No access to the file server
- No access to shared printers
- VoIP phones show “Network unavailable”
- Internet access is intermittent

Other areas of the office are functioning normally.



Step-by-Step Troubleshooting Workflow: Steps 1 to 3

Step 1: Identify the Scope	Step 2: Physical Layer Checks (Layer 1)	Step 3: Verify IP Configuration (Layer 3)
- Affected: 10 workstations + 6 VoIP phones + 1 printer - Unaffected: All devices on the west wing - Shared factor: All affected devices connect to Switch B	Use: Visual inspection, Cable tester, Switch port LEDs Check: - Are switch ports lit? - Any damaged cable? - Loose patch panel connection?	On an affected workstation: <pre>ipconfig /all</pre> Check: IP address, Subnet mask, Default gateway, DHCP lease
Conclusion: Issue likely originates at Switch B or its uplink.	Finding: Ports 1-12 on Switch B show amber/blinking error lights, indicating duplex mismatch or link instability.	Finding: Workstations show 169.254.x.x APIPA addresses → DHCP unreachable.





Step-by-Step Troubleshooting Workflow: Steps 4 to 6

Step 4: Test Connectivity	Step 5: Check Switch B Uplink	Step 6: Replace Cable and Retest
Ping the gateway <pre>ping 192.168.10.1</pre> Fails. Ping another workstation. on Switch B Fails. Ping a workstation on Switch A Fails.	Tools: • Switch web UI or CLI • Cable tester Check: • Uplink port status • VLAN configuration • Duplex/speed settings	• Replace uplink cable • Verify link lights • Ping gateway again • Check DHCP lease renewal
Conclusion: Switch B is isolated from the LAN.	Finding: Uplink port shows “Down” due to a failed Ethernet cable between Switch A and Switch B.	Result: All devices regain connectivity, VoIP phones re-register, Printers and file server accessible.





ROOT CAUSE ANALYSIS:

A damaged uplink cable between **Switch A** and **Switch B** caused:

- Loss of DHCP connectivity
- Isolation of east-wing devices
- LAN resource unavailability
- Intermittent Internet access

Tools Used

- Ping (ICMP)
- Ipconfig
- Switch management interface
- Cable tester
- ARP inspection
- DHCP less verification

FINAL SUMMARY:

A small office LAN experienced connectivity issues affecting 16 wired devices. The root cause was a damaged uplink cable between two managed switches, isolating part of the network. Troubleshooting involved physical inspection, IP configuration checks, switch diagnostics, and connectivity testing using standard LAN tools and IPv4 protocols.





PERSONAL REFLECTION:

I come to this class with prior work experience working at a large enterprise as an application support specialist who has partnered with the NOC (Networking Operations Center) to resolve incidents. For this exercise and using AI on a small scale, I found AI very helpful. I see the limitations of using AI in an enterprise environment for someone with limited network experience. In those environments there is a lot to cover. This will create a challenge for someone with limited exposure to enterprise networks. I believe that higher education, and certifications with on the job training is the best way to proceed in any career. Fungibility with a diverse skill set, including how to use AI will provide sustainability. Change is inevitable in technology, and curiosity supports innovation.



PROFESSOR FEEDBACK:



“This is a comprehensive application of generative AI to a realistic LAN troubleshooting scenario. I especially appreciate how the troubleshooting process isolates the problem rather than immediately assuming a cause. Recognizing that the affected workstations, phones, and printer all share Switch B provides an important clue, and the subsequent physical, IP configuration, and connectivity tests logically narrow the problem to the switch uplink. The 169.254.x.x addresses provide another valuable indication that affected devices cannot reach DHCP. Your reflection is equally strong because you connect this exercise with your enterprise application-support and NOC experience.”

